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Multi-"-Omics" Profiling in Patients With Quiescent Inflammatory Bowel Disease Identifies Biomarkers Predicting Relapse

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Background: Inflammatory bowel diseases (IBD) are characterized by intermittent relapses, and their course is heterogeneous and unpredictable. Our aim was to determine the ability of protein, metabolite, or microbial biomarkers to predict relapse in patients with quiescent disease.

Methods: This prospective study enrolled patients with quiescent Crohn disease and ulcerative colitis, defined as the absence of clinical symptoms (Harvey-Bradshaw Index ≤ 4, Simple Clinical Colitis Activity Index ≤ 2) and endoscopic remission within the prior year. The primary outcome was relapse within 2 years, defined as symptomatic worsening accompanied by elevated inflammatory markers resulting in a change in therapy or IBD-related hospitalization or surgery. Biomarkers were tested in a derivation cohort, and their performance was examined in an independent validation cohort.

Results: Our prospective cohort study included 164 patients with IBD (108 with Crohn disease, 56 with ulcerative colitis). Upon follow-up for a median of 1 year, 22 patients (13.4%) experienced a relapse. Three protein biomarkers (interleukin-10, glial cell line–derived neurotrophic factor, and T-cell surface glycoprotein CD8 alpha chain) and 4 metabolomic markers (propionyl-L-carnitine, carnitine, sarcosine, and sorbitol) were associated with relapse in multivariable models. Proteomic and metabolomic risk scores independently predicted relapse with a combined area under the curve of 0.83. A high proteomic risk score (odds ratio = 9.11; 95% confidence interval, 1.90-43.61) or metabolomic risk score (odds ratio = 5.79; 95% confidence interval, 1.24-27.11) independently predicted a higher risk of relapse over 2 years. Fecal metagenomics showed an increased abundance of *Proteobacteria* ($P = 0.0019$, $q = 0.019$) and *Fusobacteria* ($P = 0.0040$, $q = 0.020$) and at the species level *Lachnospiraceae_bacterium_2_1_58FAA* ($P = 0.000008$, $q = 0.0009$) among the relapses.

Conclusions: Proteomic, metabolomic, and microbial biomarkers identify a proinflammatory state in quiescent IBD that predisposes to clinical relapse.

Key Words: relapse, prediction, microbiome, metabolomics, proteomics, inflammatory bowel diseases, Crohn disease

INTRODUCTION

Inflammatory bowel diseases (IBD), comprising Crohn disease (CD) and ulcerative colitis (UC), are chronic inflammatory gastrointestinal diseases that often have an onset during young adulthood.[1] Characteristic of these diseases is a relapsing-remitting course and heterogeneity in disease behavior. Despite achieving remission, 10% to 30% of patients will experience a disease flare annually.[2, 3] Although factors such as a lack of complete mucosal healing, therapy nonadherence, and superimposed gastrointestinal infections may trigger some flares,[4] relapses are largely unpredictable. Their consequences in terms of disease-related morbidity, need for hospitalization or surgical intervention, and therapy escalation are significant. Because of clinicians' limited understanding of the reasons for relapse, it remains difficult to deliver targeted interventions to prevent relapse or reduce its impact. Having biomarkers to

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predict the future risk of relapse in patients in endoscopic remission has several benefits, important among which are therapy titration or targeted interventions before onset of symptoms that maximize the potential to remain in remission and prevent the accrual of disease-related direct and indirect costs.

Serum and fecal biochemical markers have become an important part of IBD care, allowing physicians to perform noninvasive measurements of inflammatory activity. However, the ability of such markers to predict relapse is suboptimal. Although fecal calprotectin elevation may be noted between 3 and 6 months before relapse in some patients,[5] its levels are often heterogeneous and influenced by the severity and location of inflammation.[6] Consequently, fecal calprotectin level is insufficiently predictive in many patients. In addition, routine serial stool monitoring may be challenging because of poor patient acceptability.

Driven by advances in instrumentation techniques, "-omics" research characterizing metabolites, proteins, and the microbiome have rapidly emerged as important tools to provide a deeper understanding of the biological architecture of IBD. However, their application has primarily been cross-sectional, characterizing differences between disease and health[7] or between active and inactive disease.[8] Literature on the longitudinal predictive value of such measures has been sparser, focusing on predicting disease-related complications from samples obtained at diagnosis[9] or identifying markers of response to therapy.[10] Few studies have attempted to predict relapse using microbial, metabolomic, or proteomic profiles. A recent Swiss study identified increases in the relative abundance of *Enterobacteriaceae* and *Klebsiella* in biopsy samples of patients with CD with a relapsing disease course.[11] A small study of 38 patients observed an association between endoscopic recurrence after resection and increased urinary levoglucosan.[12] However, no prior studies have examined the predictive value of metabolomic, proteomic, and metagenomic perturbations in patients with quiescent disease. Comprehensive identification of such markers may provide insights into the mechanisms of relapse in patients with IBD and lead to clinically useful tools impacting patient care. In addition, such biomarkers are an important prerequisite for the current therapeutic practice of "treat-to-target," in which early resolution of inflammation is an important step toward preventing disease progression and improving treatment outcomes.[13]

The aim of our prospective cohort study was to (1) identify and validate proteomic and metabolomic markers of disease relapse in patients with clinically and endoscopically quiescent CD or UC, and (2) to associate such changes with alterations in the gut microbiome that may predispose to intestinal inflammation.

METHODS

Study Population and Biospecimen Collection

This observational cohort study enrolled adult patients (aged ≥18 years) with quiescent CD or UC receiving care at Massachusetts General Hospital between March 2016 and December 2018. To be included, patients needed to have a confirmed diagnosis of CD or UC and have stable quiescent disease in the 1 year before this study. Quiescent IBD was defined as (1) stable lack of clinical symptoms (Harvey-Bradshaw Index[14] ≤4 for CD; Simple Clinical Colitis Activity Index[15] ≤2 for UC), (2) colonoscopy within 1 year prior showing no endoscopic inflammation (Mayo score of 0 for UC; absence of ulcers for CD), (3) normal C-reactive protein (CRP) and erythrocyte sedimentation rate (ESR), and (4) normal fecal calprotectin level at enrollment. All patients signed informed consent. Patients included in this study were enrolled either as part of an ongoing case-control study examining the mechanism of fatigue in patients with quiescent IBD[16] or from our ongoing institutional IBD registry[10] (Fig. 1). All participating subjects were invited to provide blood samples and optionally a fecal sample for metagenomic analysis.

Definition of Relapse

Our primary outcome was clinical relapse within 2 years, defined as symptomatic worsening along with elevation of CRP, ESR, or fecal calprotectin resulting in a change in medication dose, switch to a different agent, initiation of systemic steroids, or IBD-related hospitalization at surgery. For each patient who relapsed, the time between enrollment and relapse was noted. The appearance of symptoms without objective confirmation of inflammation or therapy change, or a change in treatment for reasons other than luminal disease activity (intolerance, insurance, extraintestinal symptoms), were not considered relapse. No patients intentionally de-escalated therapy at enrollment.

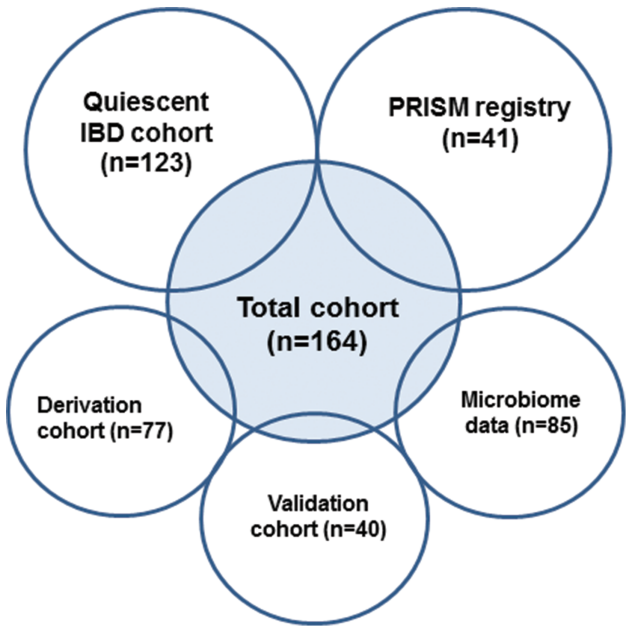


FIGURE 1. Venn diagram of included participants in the study. Abbreviation: PRISM, Prospective Registry in IBD Study at Massachusetts general hospital.

Covariates

We obtained detailed information regarding demographics and specific disease characteristics. including age at diagnosis, sex, disease extent, location and behavior according to the Montreal classification,[17] smoking history, and current and past medical and surgical treatments. We also noted baseline values of hemoglobin, white blood cell count, CRP, and ESR. We reviewed the most recent colonoscopic evaluation before enrollment and noted if patients had ongoing histologic activity.

Proteomic, Metabolomics, and Metagenomics Profiling

Collected serum samples underwent proteomic and metabolomics profiling using multiplex proximity extension assay technology and liquid chromatography-mass spectrometry methods, respectively. Fecal samples were sent to the Broad Institute (Cambridge, MA) for shotgun metagenomic sequencing on the Illumina HiSeq platform. A detailed description regarding the analysis is provided in the Supplemental Material.

Statistical Analysis

This study was approved by the institutional review board of the Partners Healthcare Human Subjects Research Committee. Clinical data were analyzed using Stata 15.0 (StataCorp, College Station, TX). First, the study cohort, comprising all patients who had metabolomic and proteomic profiles and an outcome of disease relapse, was randomly divided into two thirds constituting the derivation cohort and one third forming the validation cohort. Proteomic profiles were compared using the Wilcoxon test to detect markers that were significantly different between patients with relapse of disease and patients without relapse. The 8 most differentially distributed proteins between the groups were used in a binomial logistic regression model adjusting for age, sex, type of IBD, and body mass index (BMI). The logistic regression models were fitted using the glm() function in R. Quality control filtering, log2-transformation, and missing data imputation were performed on derived metabolomics profiles. Statistical analysis was performed with these post–quality control data sets, using a composite of nonsupervised multivariate analysis and principal component analysis and a permutation test between the 2 groups. Conditional logistic regression models for single-metabolite and metabolite network clusters were processed to collect adjusted odds ratios (ORs) and 95% confidence intervals (CIs) to assess correlation with relapse.

For the cohort, we divided proteins and metabolites significantly associated with relapse (adjusted P value <0.05) into tertiles and ranked each tertile from the lowest to the highest concentrations of the measure (Supplemental Table 1) A sum of the ranking scores was used to create separate proteomic and metabolomic risk scores to predict relapse. The predictive value of these risk scores was examined in the independent validation cohort. The accuracy of the predictive models in the derivation and validation cohorts were examined by comparing area under the receiver operating curve characteristics. The incremental value of the clinical, proteomic, and metabolomic profiles in predicting relapse was examined using likelihood ratio tests within nested multivariable models. A detailed description regarding the statistical analysis of the metagenomics data is provided in the Supplemental Material.

RESULTS

Study Population

A total of 164 patients (108 with CD, 56 with UC) were enrolled in this study with a mean age of 39.8 years (Fig. 1). Just less than half of those in the study cohort were women (44%). Upon follow-up, a total of 22 patients (13.4%; 14 with CD, 8 with UC) experienced a relapse of disease over 2 years (median time to relapse = 378 days). A total of 117 patients had available metabolome and proteome data and formed the derivation (77 patients, 10 relapses; 13%) and validation (40 patients, 3 relapses; 12%) cohorts for the serum biomarkers. The 2 cohorts were similar in demographics, disease characteristics, and current and past medical history except that patients in the validation cohort had a higher mean BMI than those in the derivation cohort (28.5 kg/m^2 vs 25.8 kg/m^2; P = 0.006) (Supplemental Tables 1 and 2).

Clinical Predictors of Relapse

Patients who relapsed were similar in demographics and most disease characteristics compared with those who remained in remission (Table 1). However, patients who remained in remission were more likely to be receiving current biologic therapy (89.6%) compared with those who relapsed (50%; P < 0.001). There was also a trend toward a greater proportion of patients without histologic activity on biopsies from their most recent colonoscopy among those who remained in remission (84.9%) compared with those who relapsed (60.0%). Clinical symptoms in the year before enrollment was similar in those who relapsed (14%) compared with those who did not (11%; P = 0.79).

Proteomics Predictors of Relapse

On multivariable logistic regression analysis adjusting for age, sex, type of IBD, BMI, and the top 8 proteins that were correlated with relapse of disease on univariate analysis, higher levels of serum interleukin (IL)-10 (P = 0.015), glial cell line–derived neurotrophic factor (GDNF; P = 0.012), and downregulated T-cell surface glycoprotein CD8 alpha chain (CD8A; P = 0.045) were significantly associated with relapse at a nominal P < 0.05 (Table 2). The serum concentrations of the 3 proteins were divided into tertiles and ranked as a tertile sum risk score, and patients were assigned a score between 3 (lowest-risk tertile for each proteomic marker) and 9 (higher-risk tertile for each marker). This risk score clearly differentiated patients who relapsed from those who remained in remission. Patients with a proteomic risk score >6 were significantly more likely to

TABLE 1. Characteristics of the Derivation Cohort of Patients With Quiescent CD or UC, Stratified by Clinical Relapse Over a 2-Year Follow-Up

Characteristic	Relapse (n = 10)	Remission (n = 67)	*P*
Age (y), mean (SD)	30.4 ± 8.8	38.7 ± 15.0	0.092
Female, n (%)	6 (60.0)	29 (43.3)	0.322
Smoking history, n (%)			0.654
Never	9 (90.0)	52 (77.6)	
Former	1 (10.0)	14 (20.9)	
Current	0 (0)	1 (1.5)	
BMI (kg/m^2), mean (SD)	24.1 ± 4.0	26.1 ± 4.7	0.200
IBD type			0.245
CD, n (%)	5 (50.0)	46 (68.7)	
UC, n (%)	5 (50.0)	21 (31.3)	
Disease duration (y), mean (SD)	11.6 ± 6.3	11.8 ± 9.6	0.944
Disease location (Montreal classification)			
CD location			0.651
L1, n (%)	1 (20.0)	12 (26.0)	
L2, n (%)	0 (0.0)	9 (19.6)	
L3, n (%)	4 (80.0)	25 (54.3)	
UC extent			0.672
E1, n (%)	0 (0.0)	2 (9.0)	
E2, n (%)	1 (20.0)	3 (13.6)	
E3, n (%)	4 (80.0)	14 (63.6)	
Perianal disease (among patients with CD), n (%)	1 (10.0)	13 (19.4)	0.472
Laboratory results, mean (SD)			
CRP (mg/L)	2.3 ± 3.6	1.9 ± 2.3	0.697
ESR (mm/h)	7 ± 6.7	9 ± 6.7	0.430
Fecal calprotectin (μg/g)	72.2 ± 38.3	37.0 ± 41.6	0.252
Interval between colonoscopy and study enrollment (d), mean (SD)	160 ± 124	166 ± 118	0.886
Complete histologic remission on colonoscopy, n (%)	6 (60.0)	56 (84.9)	0.059
Clinical remission for at least 1 year, n (%)	8 (80.0)	51 (76.1)	0.787
Past medical history			
Prior biologic therapy, n (%)	3 (30.0)	23 (34.3)	0.787
Prior surgery, n (%)	3 (30.0)	21 (31.34)	0.932
Current therapies			
5-aminosalicylic acid therapy, n (%)	5 (50.0)	13 (19.4)	0.033
Immunomodulator therapy, n (%)	4 (40.0)	23 (34.3)	0.726
Biologic therapy, n (%)	5 (50.0)	60 (89.6)	0.001

relapse (7/20, 35%) than were those with a risk score ≤6 (3/57, 5%) (log-rank $P = 0.001$) (Fig. 2A). The area under the curve (AUC) of this model was 0.75.

Metabolomics Analysis

After adjusting for age, sex, type of IBD, and BMI, whereas we observed lower levels of propionyl-L-carnitine (PLC; β, –1.24; $P = 0.009$), sarcosine (β, –0.92; $P = 0.047$), and carnitine (β, –0.95; $P = 0.015$) in patients with relapsed IBD, sorbitol was at a higher concentration (β, 1.06; $P = 0.036$). We developed a metabolomics risk score that was a sum of the ranked tertiles from highest to lowest of the 4 significant metabolites with inverse weighting for sorbitol. This score ranged from 3 to 10 (median = 0; interquartile range, 0-1). One-third of patients with a metabolomic risk score ≥6 relapsed (32%) compared with 7% of patients with a score <6 (Fig. 2B) (log-rank $P < 0.0001$). The AUC of this model was 0.70.

Combined "-Omic" Model

We then generated a multivariable model that included both the proteomic and the metabolomic risk score. This combined model yielded an AUC of 0.83, which was superior to the

metabolomic model alone ($P = 0.003$) or the proteomic model alone ($P = 0.02$). A high proteomic risk score (OR = 9.11; 95% CI, 1.90-43.61) or metabolomic risk score (OR = 5.79; 95% CI, 1.24-27.11) independently predicted a higher risk of relapse over 2 years. When adjusting for the metabolomic and proteomic markers, we found that histologic remission was no longer significantly predictive of relapse although current use of biologic therapy remained inversely associated (OR = 0.13; 95% CI, 0.02-0.83), increasing the AUC to 0.91 when it was included in the model.

Validation in an Independent Cohort

We then examined the predictive value of proteomic and metabolomic perturbations in an independent cohort of 40 patients with CD and UC. Results showed that both the proteomic risk score ($P < 0.001$) and metabolomic risk score ($P = 0.001$) were similarly significantly informative in predicting the future risk of relapse (Fig. 3).

TABLE 2. Multivariate Analysis of Significant Proteins and Metabolites Associated With Relapse in Patients With Quiescent UC or CD (derivation cohort, n = 77)

	Coefficient	*P*
Protein		
IL-10	2.36	0.015
GDNF	10.37	0.012
CD8A	–2.36	0.045
Metabolite		
PLC	–1.24	0.009
Carnitine	–0.95	0.015
Sorbitol	1.06	0.036
Sarcosine	–0.92	0.047

Derivation cohort, n = 77. Adjusted for age, sex, IBD type, and BMI.

Microbiome Analysis

Metagenomic profiles were generated for 85 patients (12 in relapse, 73 in remission) for whom stool was available. Of these, 38 patients (2 in relapse, 36 in remission) also had proteomic and metabolomic profiles. We found no significant differences in alpha- and beta-diversity as measured by the Shannon index and Bray-Curtis dissimilarity, respectively, between those who relapsed and those who remained in remission. While adjusting for age, sex, and type of IBD, we observed several significant alterations in microbial abundance at the phylum and species levels between patients with relapse of disease and patients without relapse. At the phylum level, an increased abundance of *Proteobacteria* ($P = 0.0019$; $q = 0.019$) and *Fusobacteria* ($P = 0.0040$; $q = 0.020$) were noted in patients who relapsed compared with patients in remission. No differences at the genus level were observed. At the species level, *Lachnospiraceae_bacterium_2_1_58FAA* was significantly more abundant in those who relapsed than in those who did not ($P = 0.000008$; $q = 0.0009$). Using EC libraries for functional pathway analysis, we identified that the urea carboxylase pathway was depleted (EC = 6.3.4.6; $P = 0.00014$; $q = 0.0912$) and the 4-hydroxybenzoate decarboxylase pathway was more abundant (EC = 4.1.1.61; $P = 0.0000069$; $q = 0.0912$) in patients with relapse of disease.

Among patients with all 3 "-omic" measurements, we observed a significant correlation between *L. bacterium_2_1_58FAA* and *Proteobacteria* abundances and

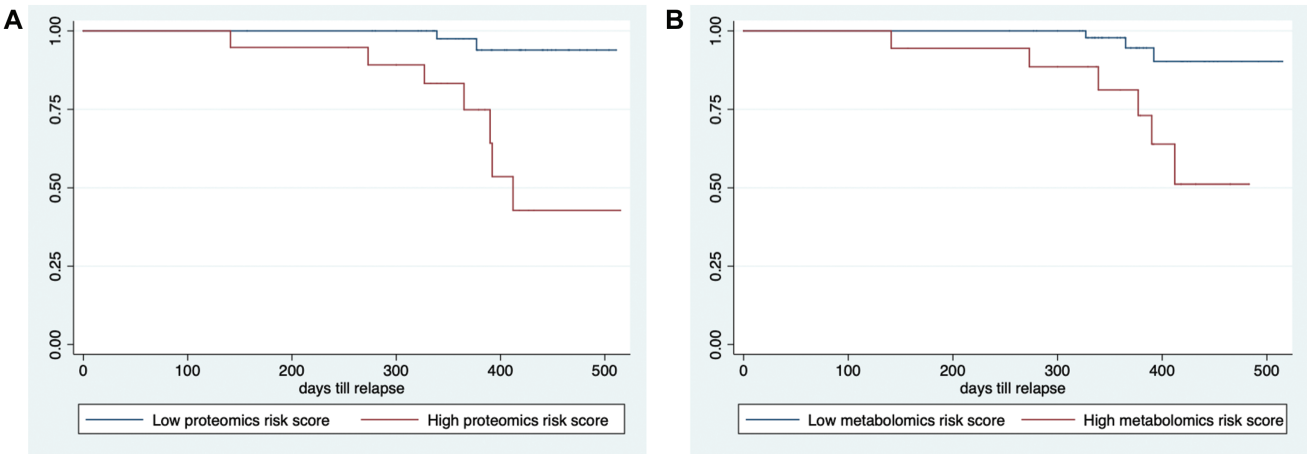


FIGURE 2. Kaplan-Meier curve describing time to relapse in patients with quiescent CD and UC, stratified by the proteomic or metabolomic biomarkers of relapse. A, Stratified by proteomics risk score. Three proteomic markers—IL-10, CD8A, and GDNF—were independently associated with relapse with $P < 0.05$ in the derivation cohort. The proteomic risk score is a sum of tertiles for each of these markers. B, Stratified by metabolomics risk score. Four metabolic markers—PLC, carnitine, sarcosine, and sorbitol—were independently associated with relapse with $P < 0.05$ in the derivation cohort. The proteomic risk score is a sum of tertiles for each of these markers.

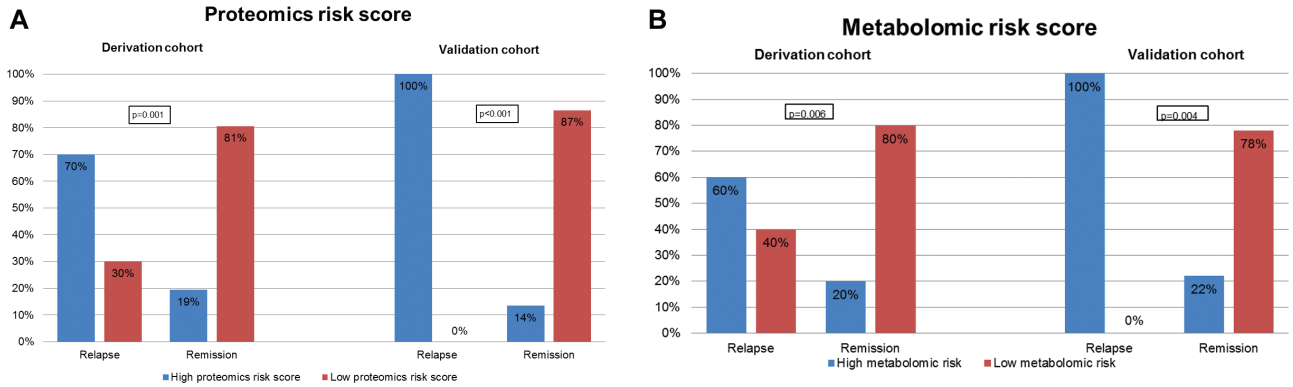


FIGURE 3. Rates of relapse among patients with high or low proteomic and metabolomic risk scores in the derivation and validation cohorts of individuals with quiescent CD or UC. A, Proteomics risk score. Definition of proteomics risk score: proteins (IL-10, CD8A, and GDNF) significantly associated with relapse (adjusted $P < 0.05$) were divided into tertiles, each of which was ranked from lowest to highest concentrations of the measure. A sum of the ranking scores was used to create the proteomics risk score to predict relapse. A cutoff >6 was defined as a high-risk score. B, Metabolomics risk score. Definition of metabolomics risk score: metabolites (PLC, carnitine, sarcosine, and sorbitol) significantly associated with relapse (adjusted $P < 0.05$) were divided into tertiles, each of which was ranked from lowest to highest concentrations of the measure. A sum of the ranking scores was used to create the metabolomics risk score to predict relapse. A cutoff >6 was defined as a high-risk score.

the proteomic risk score ($P = 0.0199$ and $P = 0.024$, respectively), with a trend toward increased abundance of those organisms with a higher metabolomic risk score ($P = 0.220$ and $P = 0.104$, respectively) (Fig. 4).

DISCUSSION

Despite being in clinical and endoscopic remission, nearly 2 out of 10 patients with IBD will relapse over the following year.[18] Such relapses are often unpredictable, and disease behavior is heterogeneous. In this prospective cohort study, we developed and validated metabolomic and proteomic biomarkers that predict the future risk of relapse in patients with quiescent IBD. Further, we showed that perturbation in these biomarkers is associated with proinflammatory changes in the gut microbiome. Such tools can be important biomarkers to monitor patients with quiescent disease and can offer mechanistic insights into the biological basis of relapse in IBD.

Among the metabolites, whereas we observed reduced levels of PLC, carnitine, and sarcosine, levels of sorbitol were elevated at baseline in those who relapsed. Functionally, several of these metabolites have been linked to biological processes relevant to intestinal inflammation. Research has shown that PLC and carnitine are important in energy metabolism and transport of fatty acids across the inner mitochondrial membrane, thus contributing to the integrity of the epithelial barrier. In addition, they play an important role in conferring resistance to oxidative stress.[19] In animal models, deficiency of carnitine has been shown to induce gut atrophy, ulcerations, and inflammation.[20] Interestingly, these metabolites have also shown promise as therapeutic interventions to ameliorate intestinal inflammation. In rats with induced severe macroscopic and histopathologic colitis, supplementation of L-carnitine resulted in improvement in histologic scores.[21] In human studies of patients with IBD on stable oral aminosalicylate or thiopurine therapy, supplementation with PLC resulted in clinical and endoscopic response in 72% and remission in 55% of mild-to-moderate patients with UC compared with 50% and 35%, respectively, in the placebo group.[22] Similar results were observed by Scioli et al,[23] who showed that a 4-week oral PLC cotreatment in patients with mild-to-moderate UC reduced mucosal inflammation and restored endothelial β-oxidation, reducing oxidative stress in the endothelium. The efficacy of PLC has been less consistent in CD.[24] Sorbitol, which was elevated in patients who relapsed, is often used as an artificial sweetener in food and is poorly absorbed in the small intestine and slowly digested by the human body.[25] Increased intake of sorbitol has been linked to gastrointestinal distress.[25]

Among the proteomic markers, we observed elevated levels of IL-10 and GDNF and depleted levels of CD8A in patients who relapsed. Although IL-10 is traditionally considered an anti-inflammatory cytokine produced by CD4+ Th cells that inhibits cytokine production by antigen-presenting cells (and consequently, one might hypothesize, should be reduced in patients who relapse), studies examining serum concentrations of IL-10 have suggested that higher levels may actually be associated with greater disease activity. In small studies, patients with CD or UC showed higher serum IL-10 levels when compared to healthy control patients, and in those with established disease, active inflammation was associated with higher serum levels of IL-10.[26] In addition, attempts to use IL-10 supplementation to treat active inflammation have yielded mostly negative results.[27] As a corollary, other immune-mediated diseases including rheumatoid arthritis also show elevated serum IL-10 levels, suggesting that this cytokine may be a biomarker to predict relapse in quiescent disease. However, the role of serological IL-10 as a biomarker in IBD remains to be elucidated. In experimental

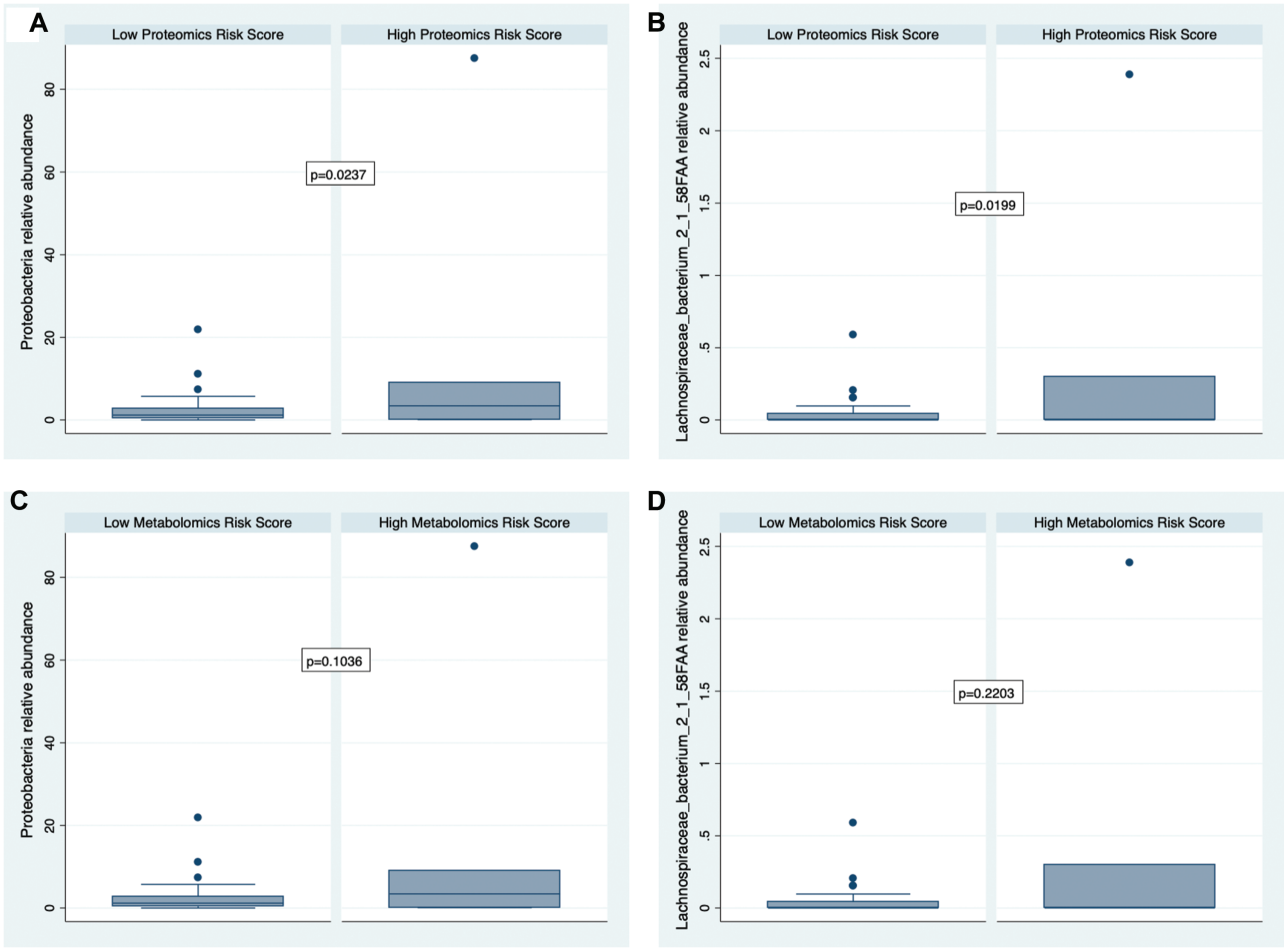


FIGURE 4. Correlation between abundance of microbial predictors of relapse and perturbations in serum proteomics (A, B) or metabolomics (C, D) in patients with quiescent CD or UC. A and B, Definition of proteomics risk score: proteins (IL-10, CD8A, and GDNF) significantly associated with relapse (adjusted $P < 0.05$) were divided into tertiles, each of which was ranked from from lowest to highest concentrations of the measure. A sum of the ranking scores was used to create a proteomics risk score to predict relapse. A cutoff >6 was defined as a high-risk score. C and D, Definition of metabolomic risk score: metabolites (PLC, carnitine, sarcosine and sorbitol) significantly associated with relapse (adjusted $P < 0.05$) were divided into tertiles, each of which was ranked from from lowest to highest concentrations of the measure. A sum of the ranking scores was used to create a metabolomics risk score to predict relapse. A cutoff >6 was defined as a high-risk score.

models, GDNF, secreted by the enteric glial cells, plays a role in maintaining epithelial barrier integrity.[28] In animal models, administration of GDNF ameliorates colitis.[28] However, GDNF itself can be induced by several proinflammatory stimuli including tumor necrosis factor-α and IL-1β. Serum levels of GDNF have not been examined in patients with IBD. We hypothesize that the elevated circulating GDNF observed in our study in those who relapsed was a consequence of such a proinflammatory state favoring relapse. The third protein, CD8A, is known to be an integral membrane glycoprotein that has an essential role in the immune response as a coreceptor for major histocompatibility complex class I molecules in T cells. Reduced CD8A levels have been seen in mice with vitamin D deficiency, resulting in increased inflammation within the gut, consistent with our association with relapsing disease.[29]

Metagenomic analysis from stool also revealed proinflammatory changes in the bacterial composition associated with the metabolomic and proteomic perturbations. *Proteobacteria* show an ability to adhere to and invade gut epithelial cells.[30] In a small cohort study of 15 children with IBD who were treated with nutritional therapy, a higher abundance of *Proteobacteria* was associated with reduced rates of sustained remission, consistent with our associations with relapse.[31] Similar associations of higher abundance of *Proteobacteria* among patients with IBD when compared with healthy control patients have also been noted.[32] We also observed an abundance of *Fusobacteria* at baseline to correlate with the risk of relapse. Although its association with colorectal cancer[33] is well established, emerging data have also linked *Fusobacterium* to intestinal inflammation.[34] A higher abundance of these phyla

was noted in postoperative patients with CD who experienced recurrence of disease.[12] A study by Forbes et al[35] detected *Fusobacteria* more frequently in inflamed CD mucosa and found an increased abundance of *Proteobacteria* in inflamed UC samples. A treatment-naïve cohort of children with CD also revealed an increased abundance of *Fusobacteria* when compared with control patients.[36]

We suggest a few key implications for our findings and directions for further study. "Treat-to-target" has emerged as the optimal approach to achieve superior outcomes in the treatment of patients with IBD, reducing future rates of relapse and preventing disease progression.[13] However, the "target" to achieve continues to evolve from clinical symptoms to endoscopic healing. Our findings suggest that despite achieving endoscopic remission (Mayo score = 0 in UC or absence of ulcers in CD), 10% to 20% of patients relapse over 2 years. We identified and validated metabolomic and proteomic biomarkers that predict such relapse and represent a subclinical proinflammatory state in such patients. Further, we associated such perturbation with a proinflammatory state in the gut microbiome, suggesting that there may be a state of deeper "'-omic' remission" that can reduce the risk of progression and even cause further relapse. Whether such deep remission is a feasible and cost-effective target needs to be established, but serum biomarkers such as those identified in our study are more attractive for routine clinical use because of wider patient acceptability than endoscopic or fecal markers. The metagenomic changes we identified also offer the possibility of altering this proinflammatory dysbiotic state through targeted microbial-directed therapies to reduce the future risk of relapse.

We readily acknowledge the limitations of our study. The sample size of our cohort, though the largest such study thus far, is moderate and relapse occurred infrequently (as would be expected in a cohort in endoscopic remission). Fecal metagenomics was available in only a subset of patients, but we were nevertheless able to identify robust associations with relapse. Further study is necessary with more dynamic sampling at different time points to identify the time interval between such "-omic" perturbations and clinical disease activity. Inclusion of time-varying measurements of environmental exposures including diet and stress are also important to better understand the mechanisms of relapse. We recognize that there are likely disease-specific differences in the mucosal immune response between CD and UC and that circulating biomarkers may be secondary to a primary process at the level of the gut mucosa. However, circulating biomarkers are more applicable to clinical care because they can be estimated repeatedly and more conveniently than sampling of the mucosal microbiome. Although all of our analyses adjusted for disease type and thus were not skewed by CD or UC distribution, larger prospective studies with sufficient power to identify disease-specific associations are needed. Finally, future studies should be adequately powered to examine the individual predictive utility of each set of biomarkers in more homogeneous cohorts such as those specifically using a single therapeutic class.

In conclusion, we showed alterations in serum metabolomics and proteomics that predicted relapse in a cohort of patients with CD or UC in endoscopic remission. Further, we validated these associations in an independent cohort. Such serum and fecal biomarkers are intriguing targets for treatment strategies to improve patient outcomes.

SUPPLEMENTARY DATA

Supplementary data are available at *Inflammatory Bowel Diseases* online.

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